

DFI command sideband ports — CS_n / CKE / ODT / parity / static

One channel, DDR5. Everything on the command bus that is *not* the CA opcode (see `opcode_rom.pdf` for the CA truth table). green =fixed level, yellow =conditional. L=0 (active-low asserted), H=1, -=no 2nd beat. **ck0**=first CK beat, **ck1**=second (2-clk commands only; 1 CK = 1 command beat, CA is SDR).

1 dfi_cs_n_pN[N_RANKS-1:0] — rank select, ONE-COLD (active-low)

The **rank** field of the phaser record decodes to a one-cold vector: the target rank's bit = L, every other rank = H. `dfi_cs_n[r] = !(r == rank)` on the select beat. CS_n is the *qualifier* — the DRAM only decodes the CA bus on the CK beat its CS_n is L. Below = the **target rank bit** per command (non-target ranks = H, except the ck1-ODT case).

command	CS_n ck0	CS_n ck1	note
ACT / MRW / WRP (2-clk, no ODT)	L	H	2-clk with no ODT control: CS_n <i>must</i> be H on ck1, else DRAM ignores the command (JEDEC note 11).
RD / WR / MRR (2-clk, ODT)	L	L / H	ck1 CS_n controls ODT on <i>non-target</i> ranks (note 8): drive L on a non-target rank to assert its ODT during the data window; H = no ODT.
PRE(ab/sb/pb), REF(ab/sb), RFM(ab/sb)	L	-	1-clk command: single select beat.
MPC / SRE / SREF / PDE / PDX / NOP	L	-	1-clk command.
DES (deselect)	H	H	all ranks H = nobody decodes the CA bus this beat.

One-cold vs one-hot: CS_n is one-cold (exactly one bit **low**). A binary `rank[clog2 N_RANKS]` → NR-wide decoder → invert → one-cold. Two different-rank commands in one bundle each drive their own rank bit low on their own phase, so a rank-0 CAS + rank-1 ACT coexist (this is the cross-rank parallelism). CKE and ODT are *not* one-hot — they are per-rank independent state (any subset may be asserted).

2 Per-rank state ports (POWER_FSM / ODT_GEN — not opcode-driven)

These do not follow the command opcode; they follow power state or the write data window. Each is a per-rank vector, driven per phase.

port	driver	value / when asserted
dfi_cke_pN[NR-1:0]	POWER_FSM	H = clock enabled (normal). Driven L for a rank on Power-Down entry (PDE) / Self-Refresh entry (SRE); held L until PDX/SRX. Per-rank independent, latency-critical (CKE setup).
dfi_dram_clk_disable_pN[NR-1:0]	POWER_FSM	H = gate CK to that rank (deep power-down / self-refresh, after CKE low). Per rank.
dfi_odt_pN[NR-1:0]	ODT_GEN	Asserted in a WL-aligned window around a WRITE (offset = odt_latency). Target and/or non-target ranks per RTT_WR/RTT_NOM/RTT_PARK. Timing-driven, not command-bit driven; overlaps the CAS by CWL.

3 Computed / static command ports

port	kind	value
dfi_parity_pN	per phase	CA-bus parity for that beat: = $\sim$ (CA0..CA13) (even parity) when parity_en ; else 0. One bit per phase, recomputed from the CA vector the encoder just built.
dfi_2n_mode	static	1 = 2N command timing (CA held 2 CK, CS pulses the sample beat — SI margin, half command rate). Config bit, boot/init only. One value all phases, no per-phase mux.
dfi_reset_n	static	L during DRAM reset, H after; driven by INIT_FSM, one-way. Not phased.

4 How each maps onto the phases (gear)

Same rule as the CA bus. Each *phased* port has one mux per phase; **phase_sel** routes the right {slot, ck-beat} value onto phase pN. At **1:1 (gear=00)** there is one phase, so a 2-clk command's ck0/ck1 land in two successive mc_clks (temporal, held by **two_clk/repeat**). At **1:2 / 1:4** the beats land on parallel phases in one mc_clk (constant spatial wiring: p0=ck0, p1=ck1, ...). **Static** ports (**2n_mode**, **reset_n**) drive one value across all phases —

no mux. A **1-clk** command uses one phase; the next phase is free for another command.